Set 2

SIMATS ENGINEERING

ASSESSMENT TOOL 2 : Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1520

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 25**
Question Count: 5

Code of Conduct

I Latisha. S (192524071) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Latisha

Assessment Tasks

Scenario-Based Questions

SET 1

1. A media company wants to migrate its video streaming platform to the cloud.
   The system must handle millions of concurrent users with minimal buffering.
   Design a cloud architecture using scalability and caching techniques.
   Explain how load balancing and auto-scaling can ensure smooth streaming.
   Discuss cost optimization strategies for such high-demand workloads.

2. A fintech startup is deploying applications using containers across multiple cloud providers.
   They face challenges in maintaining consistency and managing deployments.
   Propose a solution using container orchestration and multi-cloud strategies.
   Explain how service discovery and scaling can be handled efficiently.
   Highlight the benefits and risks of a multi-cloud deployment model.

3. An e-learning platform experiences data loss due to accidental deletion in the cloud.
   The organization needs a robust backup and disaster recovery strategy.
   Design a solution using cloud-native backup, replication, and versioning.
   Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved.
   Provide steps to ensure business continuity during failures.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data.
   Some applications must remain on-premises due to compliance requirements.
   Design a hybrid architecture that ensures secure communication between environments.
   Explain how identity management and encryption should be implemented.
   Discuss challenges in managing hybrid cloud environments.

5. A SaaS provider needs to ensure high availability and fault tolerance for its application.
   The system must continue operating even if one region goes down.
   Propose a multi-region deployment strategy in the cloud.
   Explain how data replication and failover mechanisms should be configured.
   Discuss monitoring and alerting strategies for proactive issue detection.

Scenario-Based Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

SET 2:

1.A retail company collects terabytes of customer transaction data every day from online and offline stores. The management wants to analyze purchasing patterns in near real time to improve product recommendations and inventory management.

- Design a cloud-based big data architecture using distributed storage and processing frameworks.
- Explain how batch processing and stream processing can be integrated.
- Discuss how data lakes and data warehouses can be used together.
- Recommend techniques to ensure scalability and cost efficiency.

2.A healthcare organization is migrating patient records and medical imaging data to the cloud. The solution must comply with strict privacy regulations while allowing authorized doctors to access records securely from multiple locations.

- Design a secure cloud architecture that protects sensitive healthcare data.
- Explain how encryption, identity and access management (IAM), and key management should be implemented.
- Discuss auditing, logging, and compliance monitoring strategies.
- Recommend methods to prevent unauthorized access and data breaches.

3.An online ticket booking platform experiences sudden spikes in traffic whenever popular events are announced. The company wants to reduce infrastructure management while ensuring fast response times.

- Design a serverless cloud architecture for the application.
- Explain how event-driven computing can improve scalability.
- Discuss the role of managed databases, API gateways, and message queues.
- Evaluate the advantages and limitations of serverless computing for this workload.

4.A multinational company gathers data from sales, finance, manufacturing, and customer support systems. The organization requires a centralized analytics platform with high-quality and consistent data.

- Design a cloud-based ETL/ELT pipeline for integrating data from multiple sources.
- Explain how data quality, metadata management, and governance should be maintained.
- Discuss the role of workflow orchestration and scheduling tools.
- Recommend strategies for optimizing query performance and storage costs.

5.A logistics company wants to predict delivery delays using historical shipment data, weather information, and live GPS tracking. The analytics platform must support continuous model training and real-time predictions.

- Design a cloud-based machine learning and analytics architecture.
- Explain how big data technologies can support model training and inference.
- Discuss the use of cloud-native AI/ML services for model deployment and monitoring.
- Recommend strategies for handling large-scale data ingestion, feature engineering, and model updates.

SET 3 :

1. A manufacturing company plans to migrate its legacy enterprise applications and databases from an on-premises data center to the cloud. The organization wants to minimize downtime while ensuring application performance and security.

- Design a cloud migration strategy suitable for the organization.
- Explain how different migration approaches (Rehost, Refactor, Replatform, Repurchase, Retire, Retain) can be applied.
- Discuss methods to validate application performance after migration.
- Recommend strategies to reduce migration risks and downtime.

2. A smart city project deploys thousands of IoT sensors to monitor traffic, air quality, energy consumption, and public safety. The collected data must be processed and analyzed in real time for decision-making.

- Design a cloud-based architecture for IoT data collection and analytics.
- Explain how message brokers, stream processing, and cloud storage are integrated.
- Discuss methods for handling high-volume, high-velocity sensor data.
- Recommend techniques to ensure scalability, reliability, and security.

3. A global e-commerce company wants executives to monitor sales performance, customer behavior, and regional trends through interactive dashboards. The solution must support real-time analytics and historical reporting.

- Design a cloud-based business intelligence architecture.
- Explain how data visualization tools integrate with cloud data warehouses.
- Discuss techniques for optimizing dashboard performance with large datasets.
- Recommend security measures to protect business reports and sensitive data.

4. An online gaming company experiences fluctuating workloads throughout the day. During peak hours, users report high latency and reduced application performance.

- Design a cloud monitoring and performance optimization solution.

- Explain how cloud monitoring tools, metrics, logs, and tracing help identify performance bottlenecks.
- Discuss strategies for resource optimization and capacity planning.
- Recommend methods to improve application availability while controlling operational costs.

5. A financial institution stores petabytes of customer transaction data for regulatory compliance and business analytics. The organization must retain historical records while managing storage costs efficiently.

- Design a cloud-based data lifecycle management strategy.
- Explain how data classification, retention policies, and archival storage should be implemented.
- Discuss the importance of governance, auditing, and regulatory compliance in big data environments.
- Recommend techniques for balancing data accessibility, security, and cost optimization.

1. Retail company - Big data architecture.

Cloud - based big data architecture:

Use cloud storage such as a data lake to store large amounts of transaction data. Use Apache spark for distributed processing and cloud databases for product recommend -ations and inventory management.

Batch and stream processing

* Batch processing : Analyzes historical sales data at scheduled intervals.
* Stream processing : Analyzes live transaction using tools such as Apache Kafka and spark streaming.
* Combining both provides historical insights and near - real-time updates.

Data lake and Data warehouse

* Data lake : Stores raw data in different formats
* Data warehouse : stores cleaned and structured data for reporting and business analysis.

Together, they support flexible data analysis and accurate decision making.

Scalability and cost efficiency

→ Use distributed processing and auto scaling
→ Choose pay-as-you-go cloud services
→ compress data and use suitable storage tiers
→ Delete unnecessary duplicate data.

This architecture helps the retailer analyze purchasing path quickly, improve recommendations and maintain suitable inventory levels.

2. Healthcare Organization – Secure cloud Architecture

Secure cloud architecture.

Store patient records and medical images in encrypted cloud storage. Use private networks, firewalls and secure connections to allow authorized doctors to access data.

Encryption, IAM and Key management.

- Encryption = Use AES-256 to protect data at rest and TLS to protect data in transmit
- IAM = Use role-based access control (RBAC) to allow only authorized doctors to access records
- Key management = Store encryption keys securly using a cloud key management service (KMS).

Auditing, Logging and compliance.

- Maintain logs of user access and data changes
- conduct regular security audits.
- Monitor suspicious activities and follow applicable health care privacy regulations.

Private unauthorized access

→ Enable multi-factor authentication (MFA)
→ Apply least-privilege access.
→ Use firewalls, intrusion detection and regular security updates.
→ Maintain secure backups.

(P). These measures protect patient privacy, prevent unauthorized access and ensure secure healthcare data management.

3. Online Ticket booking - serverless architecture

Serverless cloud Architecture.

Use an API gateway to receive booking requests, serverless functions such as AWS Lambda to process them, a managed database to store bookings, and a message queue to manage requests.

Event-driven computing:

Serverless functions executes when events occur, such as ticket searches or booking requests. Auto-scaling helps handle sudden traffic spikes without manually managing servers.

Role of cloud services

→ API Gateway = Receives and routes user requests.
→ Managed database = Stores ticket availability and booking details
→ Message queue = Handles requests reliably during heavy traffics.
→ Serverless functions = Process payments and bookings.

Advantages & Limitations

Advantages.

* No server management
* Automatic scaling
* Pay only for resources used
* Faster application development

Limitations.

* cold-start delays may occur
* Execution time limits may apply
* Heavy workloads can increase costs
* Vendor lock-in may occur.

4. Multinational company — cloud ETL / ELT pipeline

Cloud based ETL / ELT pipeline.

Collect data from sales, finance, manufacturing and customer support systems. Integrate the data into cloud storage or a data into cloud storage or data warehouse for analysis.

ETL : Extract → Transform → Load

ELT : Extract → Load → Transform.

Data quality, metadata and governance

* Validate data and remove duplicates.
* Maintain metadata describing data sources and formats.
* Apply data governance policies for data quality, security and access control.

Workflow orchestration and scheduling.

Tools such as Apache Airflow or cloud-native workflow services schedule and coordinate data pipelines. They manage dependencies, retries and failure notifications.

Query performance and storage costs

* Use partitioning and indexing where supported.
* Optimize queries and avoid unnecessary data scans.
* Compress data and use suitable storage tiers.
* Remove unused intermediate data.

(*). A cloud ETL / ELT pipeline integrates data from multiple departments, improves data consistency and supports reliable business analytics.

5. Logistics company - Machine Learning and Analytics.

Cloud-based ML architecture.

Collect historical shipment data, weather information and live GPS data. Store the data in cloud storage, process it using big data tools, train a machine learning model and deploy it to predict delivery delays.

Big data technologies.

- Apache spark
- Distributed storage
- Machine learning
- Real-time inference.

Cloud-native AI/ML services.

use services such as Amazon Sagemaker or Azure Machine learning to train and deploy models. Monitor model accuracy, prediction latency and data changes to maintain performance.

Data ingestion and model updates

* use message queues or streaming services to collect GPS data continuously
* Clean and transform data before training.
* Create useful features such as distance, weather and traffic conditions
* Retrain models periodically using new shipment data

⊛. A cloud-based ML system enables continuous data processing and real-time delay predictions, helping logistics companies improve delivery planning and customer satisfaction.